

Optimal Control of a Variable-Mass Thrust-Vectoring Lander

Ethan Lien

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Abstract

This document reports the attempt of solving a number of optimal control problems starting through the framework of Pontryagin's Maximum Principle. Beginning from derivation of linearized dynamics, a number of classes of optimal control problems (LQR, trajectory tracking, minimum time) are explored. Extension to nonlinear dynamics is attempted — the necessary PMP conditions are able to be derived, but numerical simulations proved to be unsuccessful. Regardless, the robustness of the linearized controllers is tested through extension to nonlinear simulation dynamics.

1 Linear-Quadratic fixed-time free-endpoint optimal control

This part of the project focuses on formulating necessary conditions for the optimal control of linearized dynamics about two cases of nominal trajectories: a trim point "hover" state and about a descending trajectory.

1.1 Linearized dynamics derivation

1.1.1 Linearization Around Hover

For the hover equilibrium, we will evaluate the dynamics about the following states:

$$v_x^* = v_z^* = \theta^* = \omega^* = 0, \quad T^* = mg$$

To begin, we start with the nonlinear dynamics equation and take partial derivatives with respect to the states and control input.

$$\dot{x} = f(x) = \begin{bmatrix} v_x \\ v_y \\ \frac{T}{m} \sin \theta \\ \frac{T}{m} \cos \theta - g \\ \omega \\ \frac{\tau}{I} \\ -\alpha T \end{bmatrix}$$

$$\partial_x f = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{T}{m} \cos \theta & 0 & -\frac{T}{m^2} \sin \theta \\ 0 & 0 & 0 & 0 & -\frac{T}{m} \sin \theta & 0 & -\frac{T}{m^2} \cos \theta \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\partial_u f = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{m} \sin \theta & 0 \\ \frac{1}{m} \cos \theta & 0 \\ 0 & 0 \\ 0 & \frac{1}{I} \\ -\alpha & 0 \end{bmatrix}$$

Next, we evaluate them at our trim point.

$$\partial_x f|_{(\bar{x}, \bar{u})} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{g}{m} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \partial_u f|_{(\bar{x}, \bar{u})} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I} \\ -\alpha & 0 \end{bmatrix}$$

To complete the linearization, we will assume m in these matrices to be m_0 . This is a sound assumption, provided that α is small, result in minimal change in state m . Thus, we can define matrices

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{g}{m_0} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_0} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I} \\ -\alpha & 0 \end{bmatrix}$$

Finally, because we have a nominal control input, we alter our definition of control signal for our linearized dynamics to be

$$u = \begin{bmatrix} \delta T \\ \tau \end{bmatrix}, \quad \delta T = T - T^* = T - mg$$

Thus, we arrive at the linearized state dynamics

$$\dot{x} = Ax + Bu$$

Using the suggested parameters provided in the problem statement, we have the following numeric values for the state matrices:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9.8100 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.4905 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0.0500 & 0 \\ 0 & 0 \\ 0 & 0.1000 \\ -0.0050 & 0 \end{bmatrix}$$

1.1.2 Linearization about descending trajectory

For the linearization about a trajectory, we will linearize about

$$\theta^* \approx 0, v_x^* \gg 0, v_z^* < 0, T^* \approx mg$$

Because this is a linearization about a trajectory, we define dynamics for state perturbation δx in terms of δu . We will continue to assume that m_0 is maintained through the trajectory. Since we want $\theta^*, T^* \neq 0$, for whatever nominal values we choose, we will have linear dynamics matrices

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{T^*}{m_0} \cos \theta^* & 0 & -\frac{T^*}{m_0} \sin \theta^* \\ 0 & 0 & 0 & 0 & -\frac{T^*}{m_0} \sin \theta^* & 0 & -\frac{T^*}{m_0} \cos \theta^* \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{m_0} \sin \theta^* & 0 \\ \frac{1}{m_0} \cos \theta^* & 0 \\ 0 & 0 \\ 0 & \frac{1}{I} \\ -\alpha & 0 \end{bmatrix}$$

I will choose $\theta^* = 0.05$ radians and $T^* = 0.95 \cdot m_0 g$. Doing so, we get the following numerical values:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9.3079 & 0 & -0.0233 \\ 0 & 0 & 0 & 0 & -0.4658 & 0 & -0.4654 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0025 & 0 \\ 0.0499 & 0 \\ 0 & 0 \\ 0 & 0.1000 \\ -0.0050 & 0 \end{bmatrix}$$

With these, we construct perturbation dynamics equation

$$\delta \dot{x} = A \delta x + B \delta u$$

1.1.3 Controllability analysis

We would like to compare the relative controllability of these two cases. The first metric we can use is their controllability matrices defined by

$$\mathcal{C} = [B \ AB \ A^2 B \ \cdots \ A^{n-1} B]$$

We can determine controllability if $\text{rank}(\mathcal{C}) = n$. For the numeric matrices we have obtained

$$\mathcal{C}_{\text{hover}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.9810 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0500 & 0 & 0.0025 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.9810 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0500 & 0 & 0.0025 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.1000 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.0050 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(\mathcal{C}_{\text{hover}}) = 7$$

$$\mathcal{C}_{\text{desc}} = \begin{bmatrix} 0 & 0 & 0.0025 & 0 & 1.1645e-04 & 0 & 0 & 0.9308 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0499 & 0 & 0.0023 & 0 & 0 & -0.0466 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0025 & 0 & 1.1645e-04 & 0 & 0 & 0.9308 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0499 & 0 & 0.0023 & 0 & 0 & -0.0466 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.1000 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.0050 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(\mathcal{C}_{\text{desc}}) = 7$$

It appears that both linearizations are controllable. To quantify this controllability, we can turn to the controllability Gramians, where a smaller minimum singular value means weaker controllability. For a finite period of time, we can use the formula

$$W_c(T) = \int_0^T e^{At} B B^T e^{A^T t} dt$$

for the finite-time controllability Gramian for our unstable system. Computing for our systems over $T = 3$ seconds:

$$W_{c,\text{hover}} = \begin{bmatrix} 8.3519 & 0 & 9.7439 & 0 & 0.7946 & 0.3311 & 0 \\ 0 & 0.0251 & 0 & 0.0130 & 0 & 0 & -0.0012 \\ 9.7439 & 0 & 11.6927 & 0 & 0.9933 & 0.4415 & 0 \\ 0 & 0.0130 & 0 & 0.0087 & 0 & 0 & -0.0008 \\ 0.7946 & 0 & 0.9933 & 0 & 0.0900 & 0.0450 & 0 \\ 0.3311 & 0 & 0.4415 & 0 & 0.0450 & 0.0300 & 0 \\ 0 & -0.0012 & 0 & -0.0008 & 0 & 0 & 0.0001 \end{bmatrix}$$

$$\min(\text{svd}(W_{c,\text{hover}})) = 4.1800 \cdot 10^{-11}$$

$$W_{c,\text{hover}} = \begin{bmatrix} 7.5188 & -0.3750 & 8.7719 & -0.4383 & 0.7539 & 0.3141 & -0.0001 \\ -0.3750 & 0.0437 & -0.4383 & 0.0348 & -0.0377 & -0.0157 & -0.0012 \\ 8.7719 & -0.4383 & 10.5263 & -0.5263 & 0.9424 & 0.4189 & -0.0000 \\ -0.4383 & 0.0348 & -0.5263 & 0.0349 & -0.0472 & -0.0210 & -0.0008 \\ 0.7539 & -0.0377 & 0.9424 & -0.0472 & 0.0900 & 0.0450 & 0 \\ 0.3141 & -0.0157 & 0.4189 & -0.0210 & 0.0450 & 0.0300 & 0 \\ -0.0001 & -0.0012 & -0.0000 & -0.0008 & 0 & 0 & 0.0001 \end{bmatrix}$$

$$\min(\text{svd}(W_{c,\text{hover}})) = 3.4293 \cdot 10^{-11}$$

We can see that the minimum singular value for the descending trajectory is lower than that of the hover case. This indicates that the descending system has weaker controllability than the hovering system. This matches the intuition that we have about the system — to correct/change the horizontal states, it is necessary to induce changes in the attitude of the lander as seen in the (3, 5) element of the descending state matrix A . Because this control scheme is so convoluted, it is quite difficult to control horizontal states for this system.

1.2 Optimal control problem

For both linearized dynamics, we want to solve

$$\min_u \int_0^{t_f} (x^T Q x + u^T R u) dt + x(t_f)^T Q_f x(t_f)$$

with fixed final time and free terminal state. To solve this, we will turn to PMP conditions to determine the form of our solution. For the finite-time, free-endpoint OCP with time-independent system dynamics, we have the PMP conditions for Hamiltonian $H(x, u, p) = \langle p, f(x, u) \rangle - \mathcal{L}$:

$$\begin{cases} \dot{x}^* = H_p|_*, & x^*(t_0) = x_0 \\ \dot{p}^* = -H_x|_*, & p^*(t_f) = -K_x(x^*(t_f)) \\ H(x^*, u^*, p^*) = \min_{u \in \mathcal{U}} H(x^*, u, p^*) \\ H^*(t) \equiv c, \quad \forall t \in [t_0, t_f] \end{cases}$$

To begin, we will start by defining our Hamiltonian and proceeding through the necessary condition.

$$\begin{aligned} H &= \langle p, f \rangle - \mathcal{L} \\ &= p^T A x + p^T B u - x^T Q x - u^T R u \end{aligned}$$

$$\begin{aligned}
H(x^*, u^*, p^*) &= \min_{u \in \mathcal{U}} H(x^*, u, p^*) \\
\implies H_u|_{u^*} &= 0 \\
H_u &= B^T p^* - 2Ru \\
0 &= B^T p^* - 2Ru^* \\
u^*(t) &= \frac{1}{2} R^{-1} B^T p^*(t)
\end{aligned}$$

This control law needs to maximize the Hamiltonian. We can determine if this is true if $H_{uu} \preceq 0$ for all $t \in [t_0, t_f]$.

$$\begin{aligned}
H_{uu} &= -2R \\
\implies R &\geq 0
\end{aligned}$$

Thus, to satisfy the necessary conditions, we must choose $R > 0$ for our cost function. Moving to the state and adjoint state:

$$\begin{aligned}
\dot{x}^* &= H_p|_* \\
&= Ax^* + \frac{1}{2} BR^{-1} B^T p^* \\
\dot{p}^* &= -H_x|_* \\
&= 2Qx^* - A^T p^*
\end{aligned}$$

Liberzon proposes and demonstrates that the adjoint solution takes the form of

$$p^*(t) = -Px^*(t), \quad \forall t \in [t_0, t_f]$$

To determine what P must be, the following is performed:

$$\begin{aligned}
p^* &= -Px^* \\
\dot{p}^* &= -P\dot{x}^* \\
2Qx^* - A^T p^* &= -2P(Ax^* + \frac{1}{2} BR^{-1} B^T p^*) \\
2Qx^* - A^T p^* &= -2PAx^* - PBR^{-1} B^T p^* \\
2Qx^* + 2A^T Px^* &= -2PAx^* + 2PBR^{-1} B^T Px^* \\
0 &= PAx^* + A^T Px^* - PBR^{-1} B^T Px^* + 2Qx^*
\end{aligned}$$

We want this to hold true for all $x^*(t)$ provided arbitrary initial state x_0 . Thus, P will be the solution to

$$PA + A^T P - PBR^{-1} B^T P + Q = 0$$

which is the continuous time Riccati equation for time-invariant parameters. Using this, we can create an optimal state-feedback law.

$$\begin{aligned}
\begin{cases} u^*(t) = \frac{1}{2} R^{-1} B^T p^*(t) \\ p^*(t) = -Px^*(t) \end{cases} \\
\implies u^*(t) = -R^{-1} B^T P x^*(t)
\end{aligned}$$

With this, we have enough to solve the two point boundary value problem with a general solution formulation, using the above ODEs and boundary conditions.

1.3 Numerical Analysis

The TPBVP can be solved using the `bvp4c()` function in Matlab, we can solve for the optimal solution provided initial conditions, final time, and cost matrices (Q, R, Q_f) .

1.3.1 Hover linearization

To demonstrate feasibility, we will start with identity cost matrices. Beginning with the hover linearization, we will use simulation parameters

$$t_f = 10, x_0 = \begin{bmatrix} 200 & 500 & 2 & 1 & 0.0017 & 0.0349 & 20 \end{bmatrix}^T$$

I find this to be a suitable choice, as we guarantee that the solution remains about the hover equilibrium point, along with the prescribed initial mass.

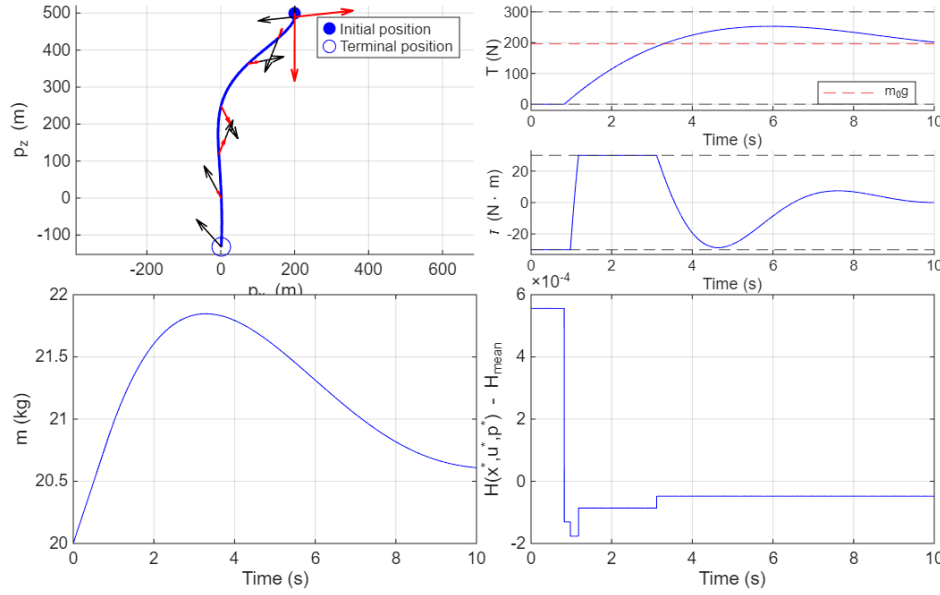


Figure 1: Hover LQR with identity cost matrices. Thrust deviation from m_0g denoted by red arrow and attitude denoted by black arrow.

This result is pretty insightful. We see that the design of the linearization drives the terminal thrust back to its equilibrium value by the end of the time horizon. Moreover, a great takeaway is the constant value of the Hamiltonian through the optimal solution of the control problem. This was exactly a necessary condition that we needed to satisfy, so this is a great indication that this solution is optimal. We even see that it remains constant in the region where the control inputs T and τ are saturated, which related to the piecewise C^1 solution that these system patterns require.

However, we do see some drawbacks of the linearization. Immediately, we can see that the lander *gains* mass throughout the trajectory, which does not match any physical intuition of the system. However, this contradiction does not break any foundations for our solution, as we know that the linearization already assumes that the mass will be relatively constant through the trajectory. Another thing to point out is the attraction of the vertical position to zero. This was something that we didn't consider in designing our equilibrium point. However, because it contributes to the cost/Lagrangian, the controller wants to drive this value to zero. This can be fixed by our choice in cost matrices by setting the entries associated with position to zero. This would mean no attraction to the origin, which is completely fine due to the free-endpoint condition of our problem. In addition, we should also be concerned that p_z goes negative — i.e. the lander goes into the ground. The solution to this, in addition to cost tuning, would be to increase the initial height of the lander.

To test this out, we can retain the same R but set $Q = Q_f = \text{diag}(\begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix})$.

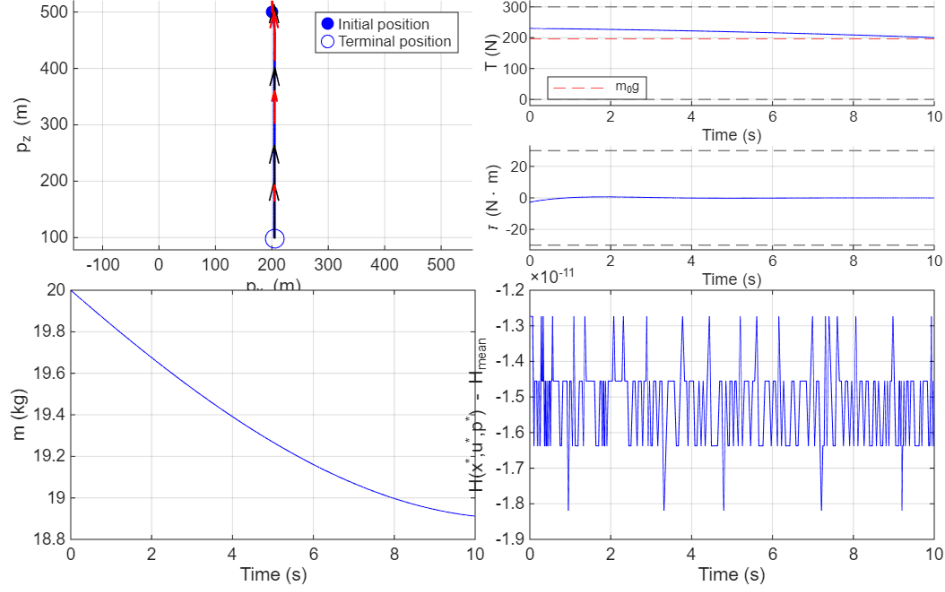


Figure 2: Hover LQR with $R = I$ and $Q = Q_f = \text{diag}(\begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix})$

This fixes the issues we wanted to address. With no draw to the origin, the lander is satisfied with hover around any positional state. Moreover, this appeared to resolve the increasing mass issue by keeping the thrust relatively constant over the trajectory. Because there is no clipping in the thrust, the Hamiltonian is also constant over the entirety of the trajectory.

1.3.2 Descending linearization

Moving on, we can observe the numerical results for the descending linearization. Because these are related to the dynamics of a perturbation, we must alter our initial condition accordingly.

$$\delta x_0 = \begin{bmatrix} 1.2 & -2.2 & 2 & 0.5 & -0.002 & 0.04 & 0 \end{bmatrix}^T$$

Key to note is that $\delta m = 0$ due to initial mass being given. We can test out the performance by once again by using identity cost matrices.

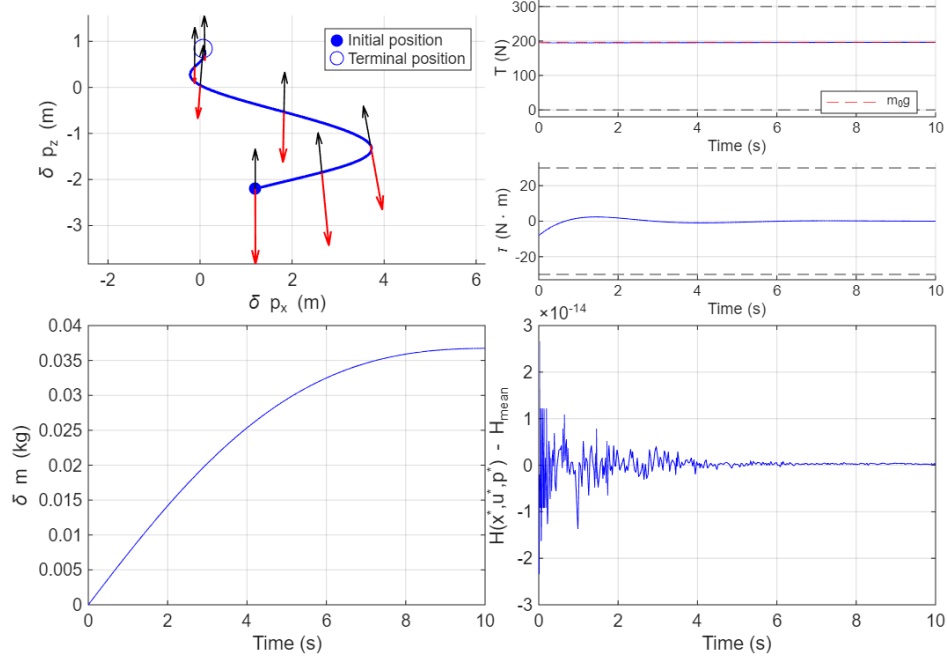


Figure 3: Descending LQR with identity cost matrices.

Due to the smaller scale of the problem, our Hamiltonian is much more constant over the interval. As we expect, there is little deviation in the control inputs. It appears that the results are similar to those produced by the hover LQR. However, because we want to control deviation dynamics, we want a different type of performance. Because we want a return to a nominal trajectory, we really desire that the perturbation state returns to the origin. Moreover, because the deviation from the control input is small, it does not need to be prioritized in the cost matrices.

Thus, we can design cost matrices: $Q = Q_f = \text{diag}([300 \ 300 \ 1 \ 1 \ 1 \ 1 \ 1])$ and $R = \text{diag}([0.5 \ 20])$.

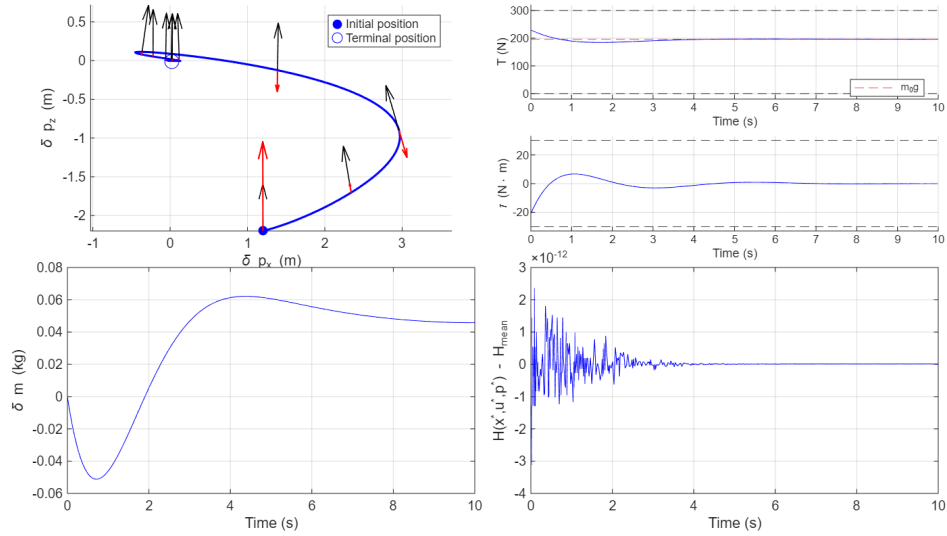


Figure 4: Descending LQR with altered cost matrices.

This performs much better than before. We see that the perturbation positions converge quickly to zero, as indicated by the terminal position and clumping of attitude markers in the position plot. The Hamiltonian

also remains constant, indicating the optimality of the controller. The caveat, though, comes in the form of increases torque input.

Now that we have obtained these two controllers, we can draw comparisons between the two. We have noticed that we have designed these controllers with different success metrics in mind. As such, these controllers are going to have different sensitivities and performances. We know that the controllers are sensitive to different initializations. When designing the hover controller, we had in mind that the terminal state was free. As a result, the controller is incredibly robust against large deviations in initial position (so long as it is chosen to not enter the floor), having similar control magnitude for a different initial states. On the other hand, the descending linearization is defined in terms of state deviations that must go to zero in a finite horizon. As a result, the further away from nominal the initial deviation is, the harder the controller must work.

With regards to control energy and transients, the design of the cost matrices greatly affect these properties. Because control effort is less of a priority for the descending case, we see stronger initial control effort to return the state deviation to zero. On the other hand, because we are looser on the penalty for positional error, the controller can distribute the control thrust over a longer period on time with lower magnitude.

2 Optimal trajectory tracking

We now want to solve a tracking problem. We want our lander to start from an arbitrary position and converge to a reference trajectory

$$x_{\text{ref}}(t)$$

in a finite amount of time.

2.1 Optimal control problem

Therefore, our optimal control problem is

$$\min_u \int_0^{t_f} [(x - x_{\text{ref}})^T Q (x - x_{\text{ref}} + u^T R u)] dt$$

Let us jump right in to the PMP necessary conditions.

$$\begin{cases} \dot{x}^* = H_p|_*, & x^*(t_0) = x_0, & x^*(t_f) = x_{\text{ref}}(t_f) \\ \dot{p}^* = -H_x|_* \\ H(x^*, u^*, p^*) = \min_{u \in \mathcal{U}} H(x^*, u, p^*) \\ H^*(t) \equiv c, \quad \forall t \in [t_0, t_f] \end{cases}$$

$$\begin{aligned} H &= p^T (Ax + Bu) - (x - x_{\text{ref}})^T Q (x - x_{\text{ref}}) - u^T R u \\ &= p^T Ax + p^T Bu - (x - x_{\text{ref}})^T Q (x - x_{\text{ref}}) - u^T R u \end{aligned}$$

$$\begin{aligned} H_u|_{u^*} &= 0 \\ H_u &= B^T p^* - 2Ru \\ 0 &= B^T p^* - 2Ru^* \\ u^*(t) &= \frac{1}{2} R^{-1} B^T p^*(t) \end{aligned}$$

$$\begin{aligned}\dot{x}^* &= H_p|_* \\ &= Ax^* + \frac{1}{2}BR^{-1}B^Tp^*\end{aligned}$$

$$\begin{aligned}\dot{p}^* &= -H_x|_* \\ &= 2Q(x - x_{\text{ref}}) - A^Tp^*\end{aligned}$$

Suppose the adjoint solution takes the form

$$p^*(t) = -2P(x^*(t) - x_{\text{ref}}(t)) + s(t), \quad \forall t \in [t_0, t_f]$$

We can determine what P must be.

$$\begin{aligned}p^* &= -2P(x^* - x_{\text{ref}}) + s \\ \dot{p}^* &= -2P(\dot{x}^* - \dot{x}_{\text{ref}}) + \dot{s} \\ 2Q(x^* - x_{\text{ref}}) - A^Tp^* &= -2P(Ax^* + \frac{1}{2}BR^{-1}B^Tp^* - \dot{x}_{\text{ref}}) + \dot{s} \\ 2Q(x^* - x_{\text{ref}}) + 2A^TP(x^* - x_{\text{ref}}) - A^Ts &= -2P(Ax^* - BR^{-1}B^TP(x^* - x_{\text{ref}}) - BR^{-1}B^Ts - \dot{x}_{\text{ref}}) + \dot{s} \\ (PA + A^TP - PBR^{-1}B^TP + Q)(x^* - x_{\text{ref}}) &= \dot{s} - A^Ts + 2P(Ax_{\text{ref}} - \dot{x}_{\text{ref}}) - PBR^{-1}B^Ts\end{aligned}$$

For all arbitrary initial condition, the left side must be zero, so we have the same Riccati equation as before. It also must be true that the right side must also be zero. Thus,

$$\dot{s} = (A - BR^{-1}B^TP)^Ts + 2P\dot{x}_{\text{ref}} - 2PAx_{\text{ref}}, \quad s(t_f) = -2Q_f x_{\text{ref}}(t_f)$$

2.2 Numerical analysis

To test the trajectory tracking, we will use the above hover trajectory as our reference. Once again, to observe feasibility, we will use identity cost matrices.

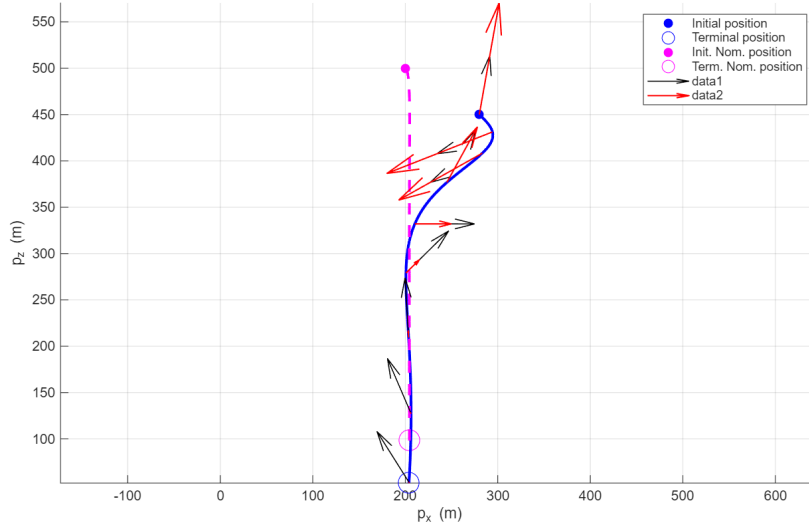


Figure 5: Trajectory tracking with identity cost matrices.

The tracking isn't the best, but it does show feasibility of the controller. It does converge to the nominal trajectory well before the end of the time interval, but there is significant overshoot in the terminal state.

After tuning, I have arrived at gains $Q = \text{diag} \left(\begin{bmatrix} 5 & 5 & 0.5 & 0.5 & 20 & 0.5 & 1 \end{bmatrix} \right)$ and $R = \text{diag} \left(\begin{bmatrix} 1 & 1 \end{bmatrix} \right)$. This choice of gains allows for decently quick convergence to the nominal state: high position gains push the trajectory to the target and low velocity gains give it the freedom to accelerate towards it.

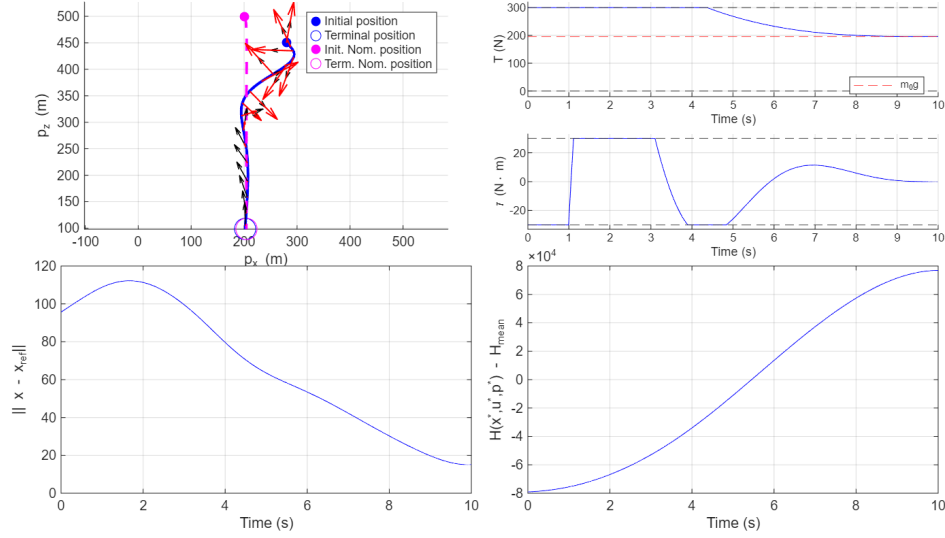


Figure 6: Trajectory tracking with tuned gain matrices.

In terms of the trajectory tracking, the performance is pretty great, with very close terminal state and quick convergence to the target trajectory. However, it is a bit concerning that we have a non-constant Hamiltonian. Something that comes to mind is that for the fixed-time fixed-endpoint OCP the adjunct state has no associated boundary conditions. As such, it would appear that it will blow up as necessary for the optimal state equation to be satisfied. The explosion would be greater for larger initial state error, which would in turn blow up the calculated Hamiltonian. We can see if this is the case by conducting trials with different magnitude initial error. The above trial had $\|x_0 - x_{ref,0}\|_2 = 95$.

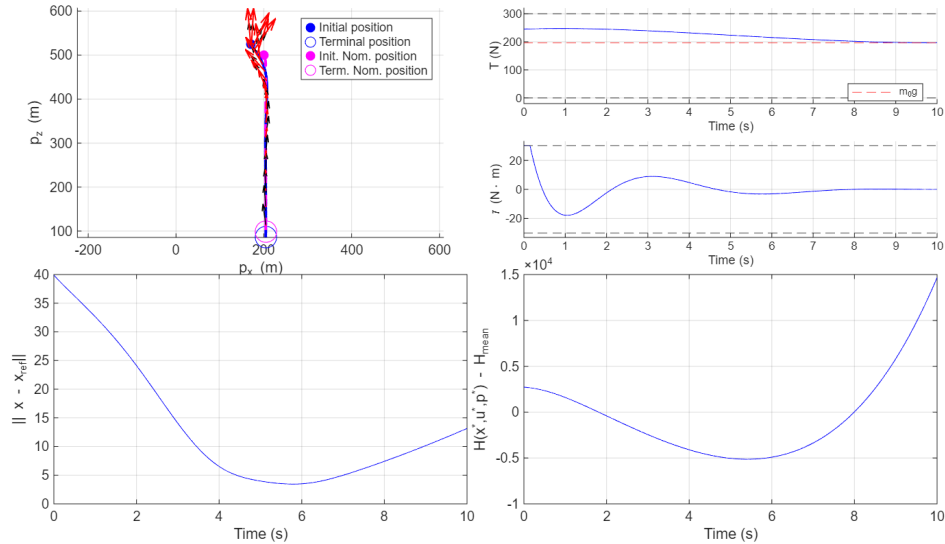


Figure 7: Trajectory tracking with $\|x_0 - x_{ref,0}\|_2 = 39.87$.

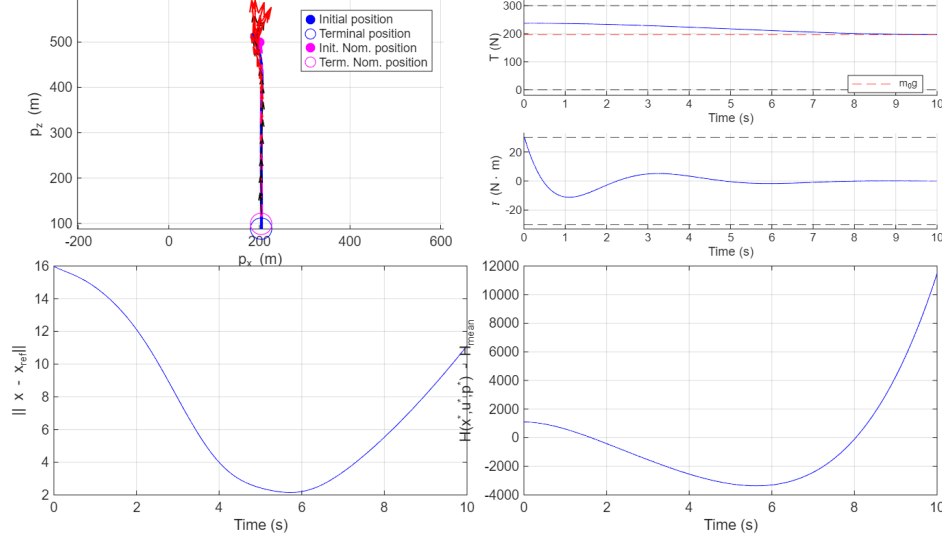


Figure 8: Trajectory tracking with $\|x_0 - x_{ref,0}\|_2 = 15.98$.

Are suspicions are *somewhat* confirmed, but not completely. We see that the the magnitude of the Hamiltonian is decreasing as we move the initial state closer to the nominal trajectory. However, it appears that it is quite significant even for very small deviation. I am not entirely sure what's going on. Perhaps there is some more nuanced with the tracking error that I have not considered. However, we do see better tracking and control performances with smaller deviation, as expected from our physical intuition.

Comparing to the previous LQR problems, we see that this behaves more as a blend between the control effort behaviors of the two cases. We see that it inherits the larger initial thrust and torque controls that the descending case had. Moreover, it continues using the large value over a longer period of time as seen in the hover case.

We can also see a little odd artifact of our gain choice in the error norm plots. Since we allowed some slackness in the velocity tracking, once the position lines up, the momentum carries in the state and the velocity continues to drive up the tracking error.

This derivation of the optimal control solution relied heavily on knowing what the reference trajectory was beforehand. We needed to know it completely to solve the adjunct state trough the time interval. If we didn't know what it was beforehand, we would need to both estimate and predict the target reference in real-time as the lander was moving. This would require knowing somewhat the dynamics that the reference will be following. If we knew this, we could use whatever reading we get of the reference to estimate the current "state" of the reference and use that in the dynamic model of the reference to predict the trajectory over a time interval. This could actually be done continuously (or at least multiple times throughout the time interval) to have better and more current predictions of the trajectory.

3 The Mission Problem

3.1 Minimum time ascent

For the ascent phase of the mission, we want to solve the optimal control problem

$$\min_u J = t_{f_1}$$

subject to linearized hover system dynamics, bounded controls, and terminal condition

$$x(t_{f_1}) = x_{sky}$$

We see that in this case the Lagrangian $\mathcal{L} = I$. We can then define the Hamiltonian.

$$\begin{aligned}
H &= \langle p, f \rangle - \mathcal{L} \\
&= p^T (Ax + Bu) - 1 \\
&\equiv 0, \forall T \in [t_0, t_f]
\end{aligned}$$

$$\begin{aligned}
\dot{p}^* &= -H_x|_* \\
&= -A^T p^* \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -g & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & \frac{g}{m_0} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1^* \\ p_2^* \\ p_3^* \\ p_4^* \\ p_5^* \\ p_6^* \\ p_7^* \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 0 \\ -p_1^* \\ -p_2^* \\ -g p_3^* \\ -p_5^* \\ \frac{g}{m_0} p_4^* \end{bmatrix}
\end{aligned}$$

Because we have a fixed endpoint, we have no conditions on the terminal state of the adjoint. Moreover, this structure indicates that some of our adjoint states are constant while others are linear in p^* .

$$\begin{aligned}
\dot{x}^* &= H_p|_* \\
&= Ax^* + Bu^*
\end{aligned}$$

This is just our state dynamics, with no dependence on the adjoint state. We have boundary conditions $x^*(0) = x_0$ and $x^*(t_f) = x_{\text{sky}}$.

$$\begin{aligned}
H &= \underbrace{p^T Ax - 1}_K + p^T Bu \\
&= K + \begin{bmatrix} p_1^* & p_2^* & p_3^* & p_4^* & p_5^* & p_6^* & p_7^* \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_0} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I} \\ -\alpha & 0 \end{bmatrix} \begin{bmatrix} \delta T \\ \tau \end{bmatrix} \\
&= K + \begin{bmatrix} \frac{1}{m_0} p_4^* - \alpha p_7^* & \frac{1}{I} p_6^* \end{bmatrix} \begin{bmatrix} \delta T \\ \tau \end{bmatrix} \\
H &= K + \left(\frac{1}{m_0} p_4^* - \alpha p_7^* \right) \delta T + \frac{1}{I} p_6^* \tau
\end{aligned}$$

Stationarity indicates that our optimal control u^* will maximize this Hamiltonian. The δT and τ that will maximize this Hamiltonian will be the following:

$$\delta T^* = \begin{cases} T_{min} - T^*, & (\frac{1}{m_0}p_4^* - \alpha p_7^*) < 0 \\ T_{max} - T^*, & (\frac{1}{m_0}p_4^* - \alpha p_7^*) > 0 \end{cases}$$

$$\tau^* = \begin{cases} \tau_{min}, & \frac{1}{I}p_6^* < 0 \\ \tau_{max}, & \frac{1}{I}p_6^* > 0 \end{cases}$$

We can observe that there does exist a singular arc where there is no defined control input. Along this arc, we have that the adjoint enters a trajectory where

$$\begin{cases} p_4^* = m_0 \alpha p_7^* \\ p_6^* = 0 \end{cases}$$

When this holds true for an interval, there is no longer a single control input that will maximize — or even affect — the Hamiltonian. We can note that this formulation is specific to the hover linearization, which assumes a constant/nominal mass. However, in reality the m_0 in the expression for p_4^* would actually be $m(t)$. Because mass will be strictly decreasing over the trajectory, the more mass is lost through thrust, the closer the system is driven towards a singular arc. This also makes sense with our physical intuition — as the mass goes to zero, so does the available thrust.

We can simulate the ascension phase with the following parameters:

$$x_{sky} = \begin{bmatrix} 0 & 300 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

$$x_0 = \begin{bmatrix} 50 & 0 & 0 & 0 & \frac{\pi}{2} & 0 & 0 \end{bmatrix}$$

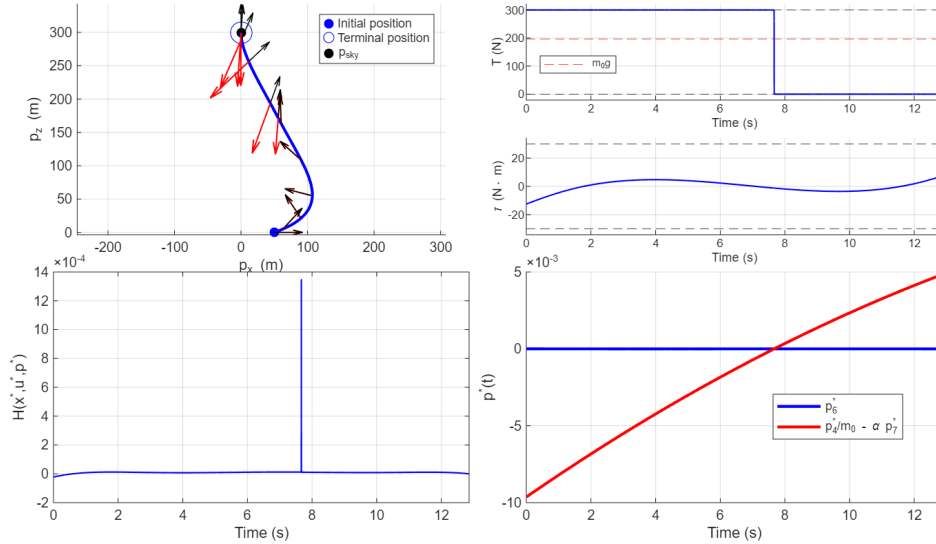


Figure 9: Ascending phase with zero terminal velocity.

This is quite nice. The lander is able to ascend to the sky state in what appears to be a low amount of time. We can observe several properties that we would expect from this controller. First, we see that $H \equiv 0$ for the entire trajectory, neglecting a small impulse at 7.5 seconds. This exactly follows the PMP condition necessary. Next, we see that our thrust input exhibits bang-bang behavior, where it initializes at its maximum thrust then uses its minimum thrust to essential coast/brake to the the terminal state. We also notice that the input torque τ does not express bang-bang behavior. This could be due to the near-zero value that its

associated adjoint state maintains through the trajectory. As such, it is more free to take on non-extreme values without affecting the maximization of the Hamiltonian. However, this does have consequence for the PMP principle. The switching of the control thrust is indicative of adjoint state ($\frac{1}{m_0}p_4^* - \alpha p_7^*$) crossing zero. Because p_6^* is also zero, there is a momentary instant where the necessary condition for existence is violated. This is reflected by in the plot of the Hamiltonian, which has a spike in value at the exact same time. However, due to the momentum/dynamics of the the adjoint states, this only happens instantaneously, and the necessary condition for existence is satisfied once more.

It would appear that the optimal strategy front-loads energy consumption, indicated by nonzero thrust occurring at the start of the trajectory. This makes sense when considering the dynamics of motion, specifically the effects of gravity. To get to an terminal position that is higher than the initial, the lander needs to overcome gravity and apply large initial thrust downward, and it is most time-efficient if it puts maximum effort into doing so. However, the requirement that the terminal velocity is zero means that the lander needs to decelerate. Because the minimum thrust is zero, it will simply cut thrust and conserve control energy by leveraging the natural free dynamics property of linear momentum to carry it to the final position. We can actually limit test this by changing the terminal velocity of x_{sky} .

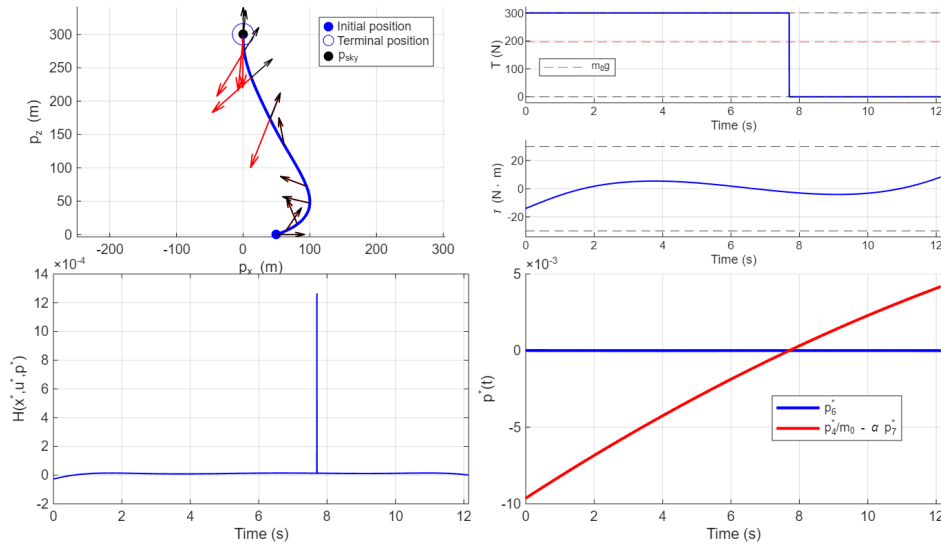


Figure 10: Ascension phase with $v_z = 8$ m/s.

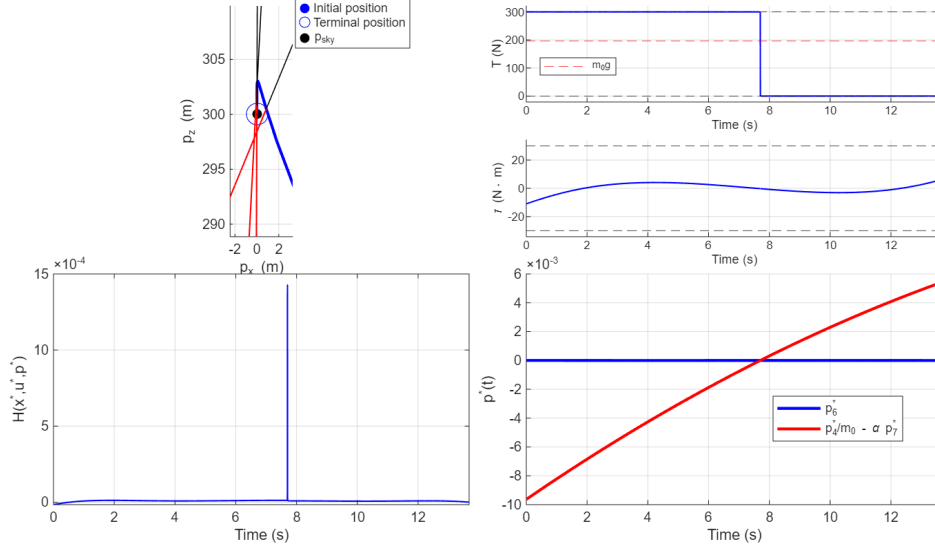


Figure 11: Ascension phase with $v_z = -8$ m/s.

The above figures demonstrate that the front-loading thrust strategy actually applies for non-zero terminal velocities as well. In both cases, the thrust is applied for a longer duration. In the positive velocity case, it uses this to accrue more momentum and have a positive velocity when it "crosses the finish line". In the negative case, however, the lander overshoots the terminal height, then uses gravity to accelerate it downwards at the terminal position¹.

3.2 Return-and-land problem

For the next phase of the mission, we want to return to a landing space while minimizing input thrust energy

$$J = \int_{t_{f1}}^{t_{f2}} (u^T R u) dt$$

with free final time t_{f2} . We will discuss two cases, in which different terminal manifolds will be considered.

3.2.1 Optimal control problem

Regardless of the manifold we study, most of the PMP conditions are actually the same. Using linearized hover dynamics,

$$\begin{aligned} H &= \langle p, f \rangle - \mathcal{L} \\ &= p^T (Ax + Bu) - u^T R u \\ &= p^T A x + p^T B u - u^T R u \\ &\equiv 0, \quad \forall t \in [t_0, t_f] \end{aligned}$$

$$\begin{aligned} \dot{x}^* &= Ax^* + Bu^* \\ x^*(0) &= x_{\text{sky}} \\ x^*(t_f) &\in \mathcal{M} = \{x \in \mathbb{R}^n : g_i(x) = 0\} \end{aligned}$$

¹ What's also really cool is that because we simply have opposite-valued terminal velocities, the lander cuts the thrust at the same time. This means that the optimal solution automatically embeds some intuitions from physics: if an object passes a certain elevation with some positive velocity, when gravity accelerates it downward it will cross that elevation at the same speed with opposite direction. Pretty neat!

$$\dot{p}^* = -A^T p$$

$$p^*(t_f) = \sum_{i=1}^{n-k} \alpha_i \nabla_x g_i|_{x^*(t_f)}$$

$$H_u = B^T p^* - 2Ru$$

$$0 = B^T p^* - 2Ru^*$$

$$u^*(t) = \frac{1}{2} R^{-1} B^T p^*(t)$$

Barring the terminal set constraint, we can use the LQR framework once again, with $Q = 0$ and $R = R$.

$$p^*(t) = -2Px^*(t)$$

where P satisfies

$$PA + A^T P - PBR^{-1}B^T P = 0$$

$$\implies u^*(t) = -R^{-1}B^T Px^*(t)$$

We can observe that the solution is nearly identical to that of the LQR, with only difference being a partial freedom in terminal state brings partial freedom in terminal adjunct state.

3.2.2 Ground landing

In this case, we want to land on flat ground with vertical touchdown and upright attitude, described by terminal manifold

$$\mathcal{M}_1 = \{p_z = 0, v_z = 0, \theta = 0\}$$

We use these to determine properties about the terminal adjunct state.

$$\nabla_x(p_z) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

$$\nabla_x(v_z) = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}^T$$

$$\nabla_x(\theta) = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}^T$$

$$\implies p^*(t_f) = \begin{bmatrix} 0 \\ \alpha_1 \\ 0 \\ \alpha_2 \\ \alpha_3 \\ 0 \\ 0 \end{bmatrix}, \alpha_i \in \mathbb{R}$$

To test feasibility, we use $R = I$ as our initial gain matrix. Continuing the mission from the zero velocity sky state, we can perform a numeric simulation.

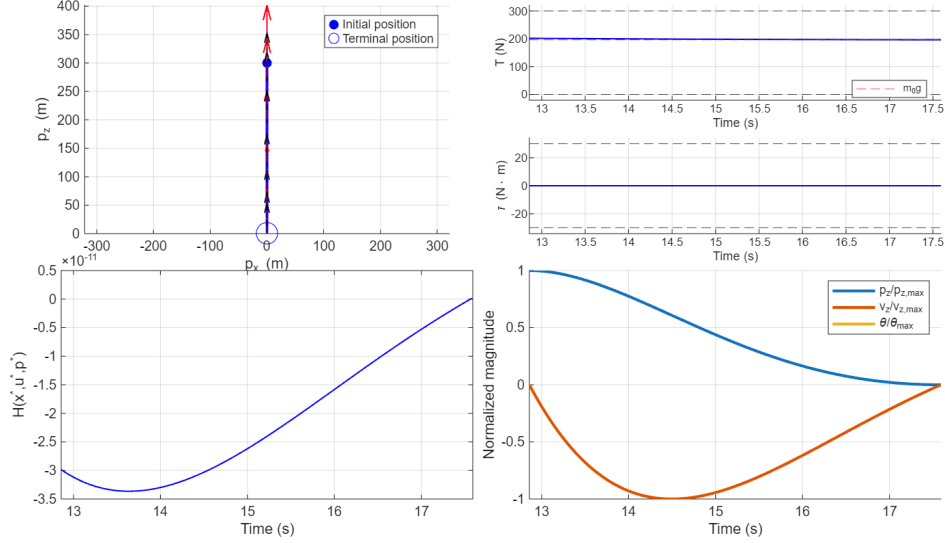


Figure 12: Ground landing with identity gain matrix.

These results demonstrate satisfaction of our problem statement and PMP conditions. We can observe that the thrust and torque commands hardly deviate from their nominal value, which is exactly what our cost function aims to reduce. Moreover, we see that the reach our target manifold at the end of the trajectory — another goal for the optimization problem. The Hamiltonian remains close to zero throughout the whole trajectory, which matches the necessary condition from PMP.

For full visualization, here is an example of both phases of the mission plotted sequentially:

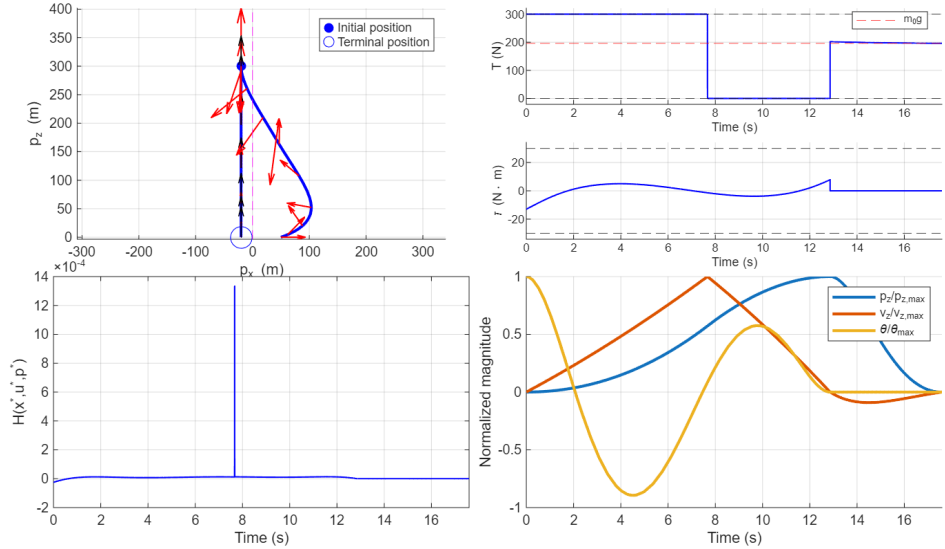


Figure 13: Full mission with ground landing.

This is just a pretty cool visualization to see. Putting this together, we see all of the previously described behavior — bang-bang control scheme for minimum time ascension, minimal control deviation for ground landing. Moreover, we also see an overarching continuity for the mission, where a completely free final time results in a zero-values Hamiltonian for the whole mission.

3.2.3 Circular platform

For this situation, we define our target manifold as

$$\mathcal{M}_2 = \{h_1(x) = 0, v_z = 0, \theta = 0\}$$

where

$$h_1(x) = (p_x - p_c)^2 + p_z^2 - r^2$$

for some horizontal center position p_c and radius r . What changes for this problem is the terminal adjoint state conditions, as we now have a different set of equations defining our manifold in $\mathbb{R}^n$.

$$\begin{aligned} \nabla_x(v_z) &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}^T \\ \nabla_x(\theta) &= \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}^T \\ \nabla_x(h_1(x)) &= \begin{bmatrix} 2(p_x - p_c) & 2p_z & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \\ \Rightarrow p^*(t_f) &- \begin{bmatrix} \alpha_1(p_x^*(t_f) - p_c) \\ \alpha_2 p_z^*(t_f) \\ 0 \\ \alpha_3 \\ \alpha_4 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

While running numerical solutions to this problem, a problem that I encountered was the lander completely running out of fuel for the mission. The root cause of this was the choice of an unreasonable target elevation for x_{sky} . As such, subsequent missions will use this sky state instead:

$$x_{\text{sky}} = \begin{bmatrix} -20 & 150 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

Moreover, numerical issues appeared with singular Jacobian error when using `bvp4c()` in Matlab. As such, the platform landing phase will use `fconmin()` instead. With identity cost matrix R and $(p_c, r) = (-40, 20)$ meters, we ge the following numerical results.

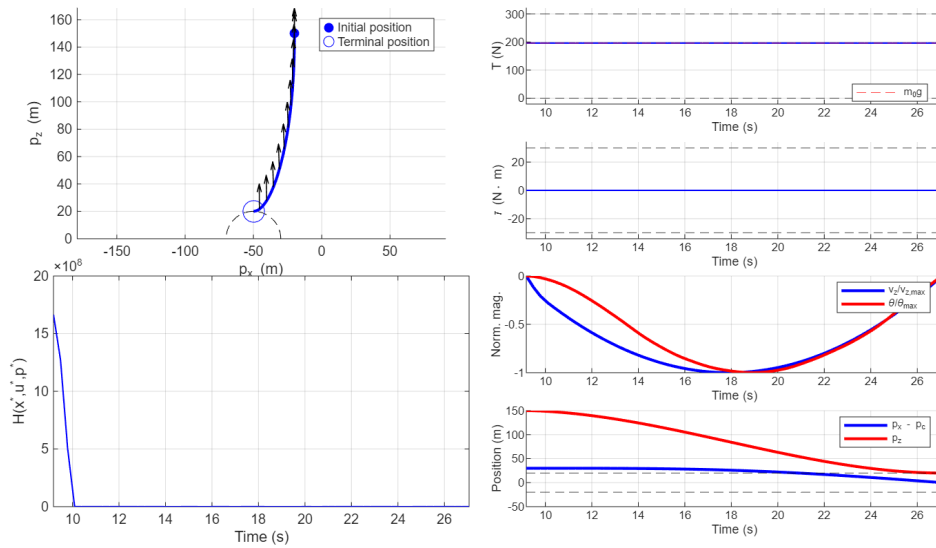


Figure 14: Platform landing with identity gain matrices

We see that the lander successfully lands on the platform with zero vertical velocity and vertical attitude. Moreover, we can see that its final position lies on the surface of the platform. As with the ground landing, there is minimal deviation from nominal thrust. This is logical, as we are using the same Lagrangian in both cases. Something that we can notice, though, is that the Hamiltonian has a harder time staying at zero during the trajectory. This may be due to the numerical implementation of `fmincon()`, which struggles to wrangle the adjunct state as it propagates backwards — it is subject to unstable dynamics. As such, we have satisfaction of the adjunct and Hamiltonian at the terminal time, but as we approach the initial time, it eventually blows up.

A full mission example is demonstrated below.

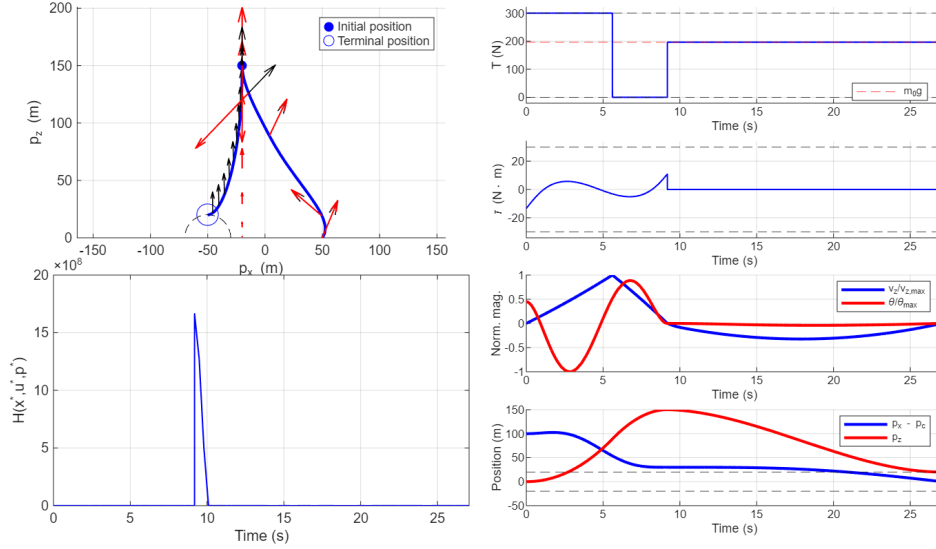


Figure 15: Full mission with platform landing.

4 Nonlinear optimal control

Now we want to utilize the full nonlinear model to maintain control about a a hover. This means our scope will be around moderate attitude angles, short durations and near-zero velocities.

4.1 Optimal control problem

We want to consider the fixed-time, free-endpoint nonlinear optimal control problem

$$\min_u \int_0^{t_f} (x^T Q x + u^T R u) dt$$

For this type of problem, we have the following PMP conditions:

$$\begin{cases} \dot{x}^* = H_p|_*, & x^*(0) = x_0 \\ \dot{p}^* = -H_x|_*, & p^*(t_f) = -K_x(x^*(t_f)) \\ H(x^*, u^*, p^*) = \min_{u \in \mathcal{U}} H(x^*, u, p^*) \\ H^*(t_f) \equiv c, \quad \forall t \in [t_0, t_f] \end{cases}$$

Beginning with our Hamiltonian,

$$\begin{aligned}
H &= \langle p, f \rangle - \mathcal{L} \\
&= p^T \cdot \underbrace{\begin{bmatrix} v_x \\ v_z \\ \frac{T}{m} \sin \theta \\ \frac{T}{m} \cos \theta \\ \omega \\ \frac{\tau}{I} \\ -\alpha T \end{bmatrix}}_{\triangleq G} - x^T Q x - u^T R u \\
&= p_1 v_x + p_2 v_z + p_3 \frac{T}{m} \sin \theta + p_4 \frac{T}{m} \cos \theta + p_5 \omega + p_6 \frac{\tau}{I} - p_7 \alpha T - x^T Q x - u^T R u
\end{aligned}$$

The optimal state dynamics follow the system dynamics, as resolved in the following:

$$\dot{x}^T = \begin{bmatrix} v_x^* \\ v_z^* \\ \frac{T^*}{m^*} \sin \theta^* \\ \frac{T^*}{m^*} \cos \theta^* \\ \omega^* \\ \frac{\tau^*}{I} \\ -\alpha T^* \end{bmatrix}$$

We can also derive the costate dynamics.

$$\dot{p}^* = - \begin{bmatrix} 0 \\ 0 \\ p_1^* \\ p_2^* \\ \frac{T^*}{m^*} (p_3^* \cos \theta^* - p_4^* \sin \theta^*) \\ p_5 \\ -\frac{T^*}{(m^*)^2} (p_3^* \sin \theta^* + p_4^* \cos \theta^*) \end{bmatrix} + 2Qx^*$$

The form of the optimal control resolves from the stationarity condition.

$$\begin{aligned}
H_u &= \left[\frac{1}{m} (p_3 \sin \theta + p_4 \cos \theta) - p_7 \alpha \quad p_6 \frac{1}{I} \right] - 2Ru \\
0 &= \left[\frac{1}{m^*} (p_3^* \sin \theta^* + p_4^* \cos \theta^*) - p_7^* \alpha \quad p_6^* \frac{1}{I} \right] - 2Ru^* \\
\implies u^* &= \frac{1}{2} R^{-1} \left[\frac{1}{m^*} (p_3^* \sin \theta^* + p_4^* \cos \theta^*) - p_7^* \alpha \quad p_6^* \frac{1}{I} \right]
\end{aligned}$$

Because there is no terminal cost, this condition, in addition to the free-endpoint condition resolves into

$$p^*(t_f) = 0$$

$$x^*(0) = x_0$$

I have had deep troubles trying to find an optimal numerical solution. Focusing on the hover case in Part I, I have attempted to use Direct Collocation but to no avail. There was always some sort of discrepancy between feasibility and optimality, where my solver could only enforce dynamically feasible solutions, but could never get any that demonstrate optimality by satisfaction of the PMP principles, or even satisfaction of terminal conditions.

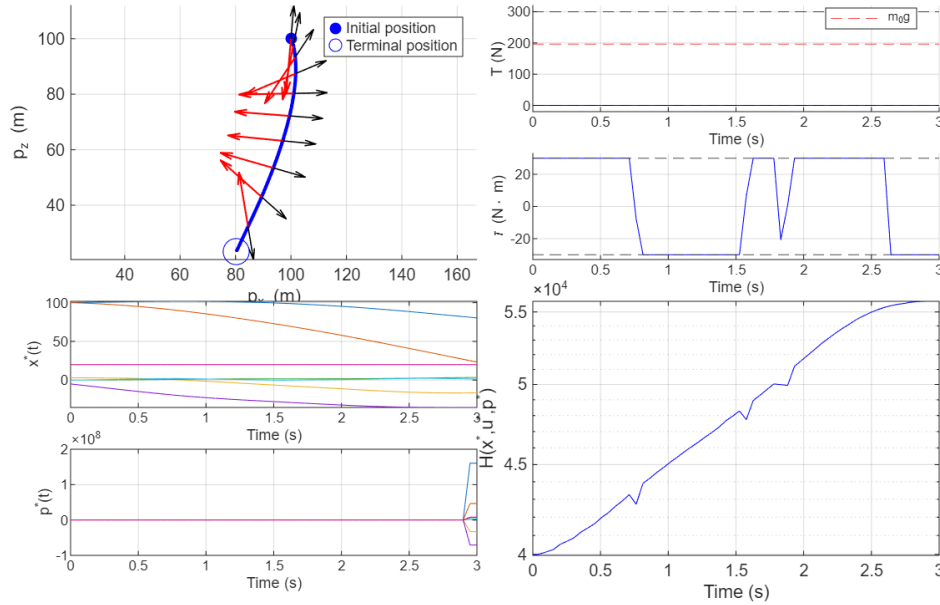


Figure 16: Example of failure by nonlinear solver

The example above demonstrates the extent to which I could get my solver to work. Looking at the trajectory it produces, it definitely seems like a feasible trajectory that the system could produce from initial conditions and a controller. However, everything else falls apart from there. Our expectation is that the Hamiltonian will be constant over the trajectory. However, we see that it grows by over 45,000 throughout the simulation. Moreover, we prescribe a terminal condition on our optimal costate $p^*(t)$, we see that it jumps to a non-zero value exactly at the end of the simulation. Attempting to debug this has been a great burden and cause of suffering even with/especially with the use of generative AI agents (Claude, ChatGPT, Gemini). No good solution was able to be found, at least with the methods and ideas that were available to me.

Though I cannot perform direct analysis of the nonlinear optimal control, at the very least I can demonstrate analysis of the linear controller with nonlinear dynamics.

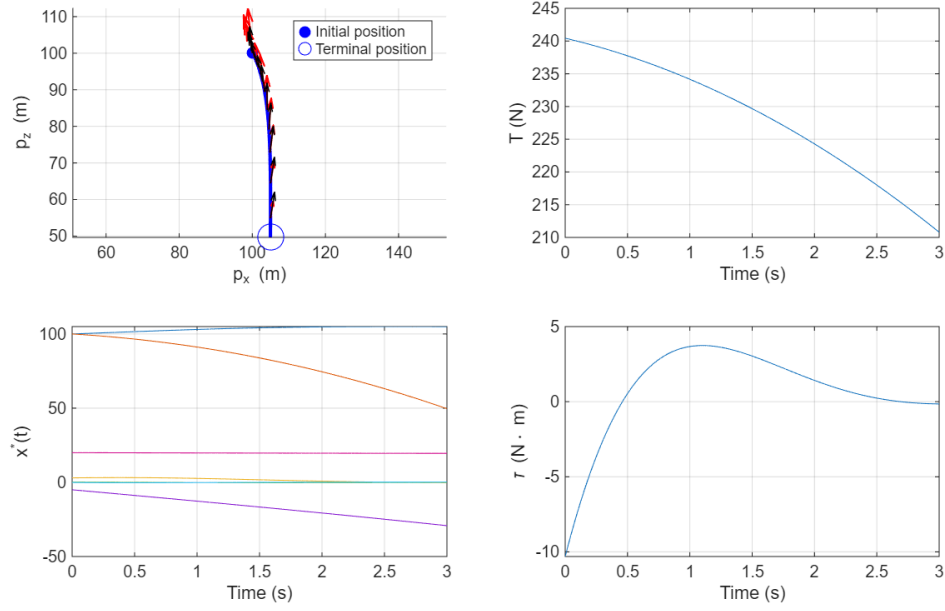


Figure 17: Linear controller acting on linear dynamics.

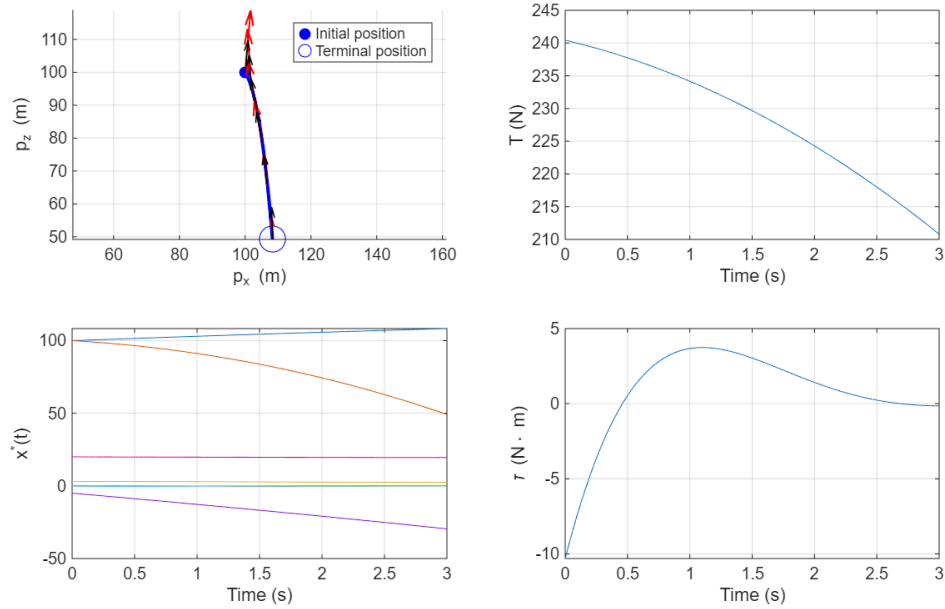


Figure 18: Linear controller acting on nonlinear dynamics.

We can see that the performance of the linear controller behaves quite well on the nonlinear dynamics. This speaks to the accuracy of the linearization about the hover equilibrium (or at least within this small deviation from the trim point). We can see that our assumption that the mass of the lander is constant holds generally true for this duration of simulation.